

Texas A&M University	College Station, TX
Pursuing M.S. Computer Science	Expected 2028
Texas A&M University	College Station, TX
B.S. Computer Science & B.S. Statistics, <i>Summa Cum Laude</i>	May 2026 GPA: 3.92/4.00

Technical Skills
Data Science & Statistics: Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, Bayesian inference, MCMC, hypothesis testing, FDR correction, Matplotlib
Bioinformatics & HPC: GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA pipelines, Slurm, Linux, TAMU HPRC
Machine Learning & Data Systems: PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, Git, GitHub Actions
Programming: Java, C++, TypeScript, JavaScript, React, FastAPI, Docker, PostgreSQL

Experience

Graduate Assistant Research (GAR) - Computational Genomics	June 2026 - Present
Ioerger Lab, Texas A&M University	College Station, TX
- Built Python pipelines for Bayesian dN/dS analysis with GenomeMap across <i>M. tuberculosis</i> lineages L1-L4. - Developed a Bayesian comparison pipeline using DPD (difference in posterior distributions) to estimate the probability of higher gene-level dN/dS in one cohort than the other, comparing 922 clinical isolates (178 from patients with diabetes; 744 without). - Corroborated selection findings through PAML/codeml and pN/pS analyses, obtaining concordant results across all three methods. - Modeled gene-level fitness under rifampicin treatment using TnSeq data, linear models, and likelihood-ratio tests.	

Software Engineering Capstone - Amazon-Sponsored	August 2025 - December 2025
Texas A&M University	College Station, TX
- Built a breach-intelligence analytics platform in a 3-person team, implementing Scrapy and Tor ingestion plus a LangChain and Gemini 2.5 Flash Lite pipeline for entity extraction and threat classification. - Designed storage and retrieval across AWS S3, MongoDB, Elasticsearch, Redis, and Neptune; delivered Streamlit and Kibana views for querying unstructured breach data by sector and geography.	

Undergraduate Researcher	January 2025 - May 2025
UrbanResilience.AI Lab, Texas A&M University	College Station, TX
Built Python ingestion and preprocessing pipelines integrating air-quality sensor APIs and web-scraped wildfire data across Los Angeles sources for downstream emergency-response analysis.	

AI Engineering Intern	May 2024 - August 2024
Videomagic	Remote
- Developed and evaluated PyTorch and Hugging Face Transformers computer-vision workflows for AI-generated video detection, analyzing frame-level outputs and model failure modes for product deployment. - Built Feathers.js, MySQL, and Knex.js APIs for model inference and data management, secured with Auth0 and JWT RBAC; shipped AI features associated with 15% higher engagement.	

Projects & Competitions

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Manuscripts in Preparation

Dave, H., Shatby, A., Smith, M., and Ioerger, T. Differential Bayesian analysis of genomic sequences identifies *M. tuberculosis* genes under positive selection in TB patients with diabetes.